



Touchless Sense Switch To Combat Covid-19

Due to the outbreak of Covid-19 pandemic, a new technology called touchless sense switch (TSS) has been introduced, which helps read human inputs without touching any equipment or switch panel. Each button/switch that needs human input through physical touch can be replaced with a sensing area activated by TSS whose broad designing process is described here



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Necessity is the mother of all inventions. And so is this new technology—touchless sense switch (TSS). The latest challenges faced by people around the globe due to the outbreak of Covid-19 pandemic have compelled new means of human-machine interactions. This means not only maintaining safe distance from commonly/publicly used switch panels but also controlling these without having any physical contact.

Touchless sense switch is designed to read human inputs without requiring a person to touch any equipment or switch panel.

Each button/switch that needs human input through physical touch can be replaced with a sensing area activated by TSS. A finger or a pointing object held at close proximity ($\sim 1\text{cm}$) above the sensing area of the switch to be operated for a pre-defined period (for example, till red LED stays on for few seconds at least) would result in activation/deactivation of the switch. This technique/methodology has several advantages that are described later in this article.

Problem statement

Numerous consoles exist at various public locations or establishments that are operated by the public or employees/workers on a large scale. These consoles take input in the form of physical human touch (tapping pads or pressing buttons) from numerous users many times in a day. The following are a few of the equipment/devices that fall under this category:

- Elevators

- Public telephones
- Coffee vending machines
- Train ticket vending machine
- Keypads on ATM machines
- Other applications in home automation

The following are the limitations or cons of existing (electro-mechanical pressing or touch-based) switches:

- These switches should not be used in certain pandemic situations such as Covid-19, where the virus can spread by just a touch/press of switches on any equipment or device placed in public areas.
- These need maintenance and regular cleaning.
- Electro-mechanical switches are prone to wear and tear and need replacement frequently.

Solution

Requirements. The following are the requirements as a solution to the above problem statement:

- A new technique/technology is required to read human inputs (human-machine interface) without the need to touch a pad or press a button.
- Must replicate the existing model without any dependency or addition of another entity by the user.
- Simple and seamless integration into the hardware architecture of the existing or new system.
- Behaviour of switches (pushbutton, on-off one-way, two-way, etc) can be set in the form of configuration parameters—all modules to remain the same, only configuration settings will alter the behaviour.
- Must require less or zero maintenance.
- Change in the behaviour of the new